

IE Capstone 2026 — Project Proposal

AI System to Catch Illegal Airbnb Listings

KPMG Spain

Lighthouse Practice

2026

THE PROBLEM

Airbnb has transformed tourism — but in cities like Paris, Málaga, Rome and Barcelona, it has also made housing unaffordable for the people who actually live there. Thousands of apartments that should be homes for local families are being rented out illegally to tourists.

City regulators know this is happening but there are not enough inspectors. Meanwhile, landlords find clever workarounds: one person registers dozens of apartments under different names, or they label tourist flats as "seasonal rentals" to avoid the rules.

Four options:

FR **Paris**

Courts have thrown out regulators' evidence because screenshots are not enough proof. The city knows who the illegal hosts are — it just cannot build a case against them.

~75,000 listings active. 31.7% of hosts operate multiple apartments, hiding commercial businesses behind personal accounts.

ES **Málaga**

Tourism has overwhelmed this city. More than half of all tourist beds are now Airbnb-style flats. Some low-income workers have been forced to live in their cars.

Short Term Rentals is 56% of all visitor accommodation. Tourist beds up 36% in just two years.

IT **Rome**

Two thirds of Airbnb listings in Rome do not even display a registration number — which is required by law. Airbnb does not check this itself, so regulators have to investigate every single complaint manually.

Only 34% of listings show a valid licence number.

ES **Barcelona**

Barcelona has already decided to ban all tourist apartments by 2028 — but thousands of illegal listings are still operating today. Rents have gone up 68% since 2015. T

10,101 licensed apartments expiring 2028. Rents up 68% since 2015.

THE SOLUTION

We are building an AI system made up of six specialised agents. Together they can scan every listing in a city, check it against the law, look at its photos for fake or misleading content, decide how serious the violation is, and write up a formal case file. A human regulator always has the final say.

THE 6 AGENTS

01 **Detection Agent**

What it does: Scans every listing

Goes through all listings looking for red flags: apartments listed 365 days a year, one person owning 20 listings, missing licence numbers, prices that make no sense for a private home.

02 **Compliance Agent**

What it does: Checks the law

Looks up the exact legal rules that apply to each flagged listing and checks whether the listing breaks them.

03 **Image Agent**

What it does: Analyses the photos

Looks at the listing's photos and description together. If someone claims 'sea views' but the photos don't, or if the images appear to be stock photos rather than the real apartment, this agent flags it.

04 **Risk Scoring Agent**

What it does: Scores each case

Takes everything the first three agents found and gives each listing a score from 0–100 based on how serious and provable the violation is. This tells regulators which cases to prioritise.

05 **Orchestrator Agent**

What it does: Runs the whole team

Decides which agent runs when, makes sure everything happens in the right order, and collects all the findings into one complete package before passing it on to the final agent.

06 **Report Agent**

What it does: Writes the case file

Produces a formal enforcement brief for the regulator: which listing, what the violation is, which law was broken, what the evidence is, and what action is recommended.